PHYC10009 Foundations of Physics
Lecture 13
WAVES

13

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THE UNIVERSITY OF MELBOURNE

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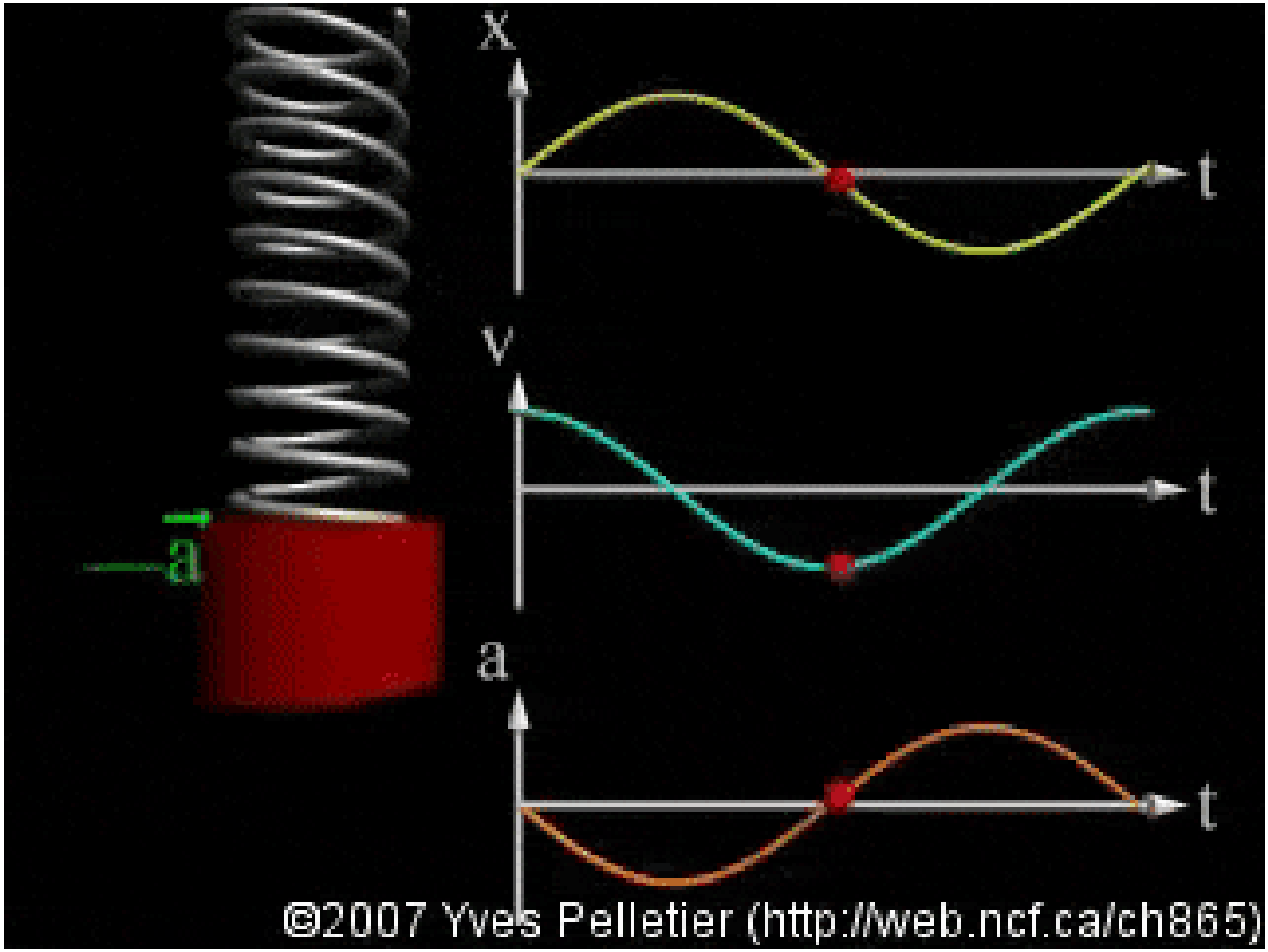
Watch the Animation



Recap

Generic SHM: $x(t) = A\cos\left(\frac{2\pi t}{T}\right) = A\cos\left(\frac{2\pi t}{T}\right) = A\cos\phi$

Angular frequency ω [rad/s] $\omega = \sqrt{\frac{k}{m}}$

Frequency $f = \frac{\omega}{2\pi}$ [Hz]

Amplitude A [units depend on application!]

Period $T = \frac{2\pi}{\omega}$ [s]

Phase ϕ *(the argument of the sinusoidal function)*

Measuring π

$$T = \frac{2\pi}{\omega} \Rightarrow \omega = \frac{2\pi}{T} \text{ and } \omega = \sqrt{\frac{k}{m}} \Rightarrow \frac{2\pi}{T} = \sqrt{\frac{k}{m}} \Rightarrow \boldsymbol{\pi} = \frac{\boldsymbol{T}}{\boldsymbol{2}}\sqrt{\frac{\boldsymbol{k}}{\boldsymbol{m}}}$$

Measure period, Spring Constant and mass for SHM can determine, $\boldsymbol{\pi}$

Data from Friday's SHM demo determines $\boldsymbol{\pi}$ with 2% precision

Measuring π

Data from in-Lecture Demo on Friday:

Mass (kg)	Length (cm)	Extension (m)	Weight (N)	K (N/m)
1	51.5	0		
2	59.7	8.2	9.8	119.5122
3	68.7	17.2	19.6	113.9535
4	77.4	25.9	29.4	113.5135
5	86.3	34.8	39.2	112.6437
6	94.5	43	49	113.9535
Average				114.7153

Time 10 cycles	Period	π
5.98	0.600	3.20
8.36	0.836	3.16
10.20	1.020	3.16
11.98	1.200	3.21

Waves

- Simple Harmonic Motion:
  lots of oscillation not going anywhere

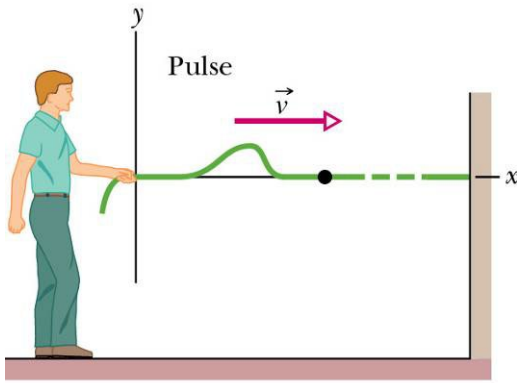
- A wave is a disturbance that propagates

- A wave is a disturbance that transports energy without transporting matter

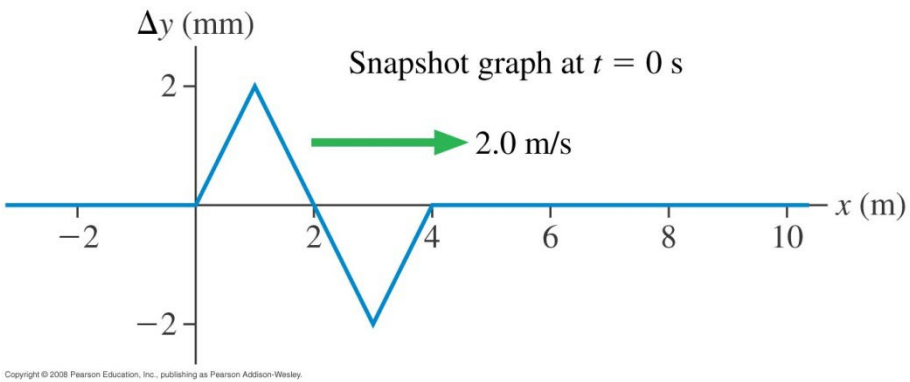
- Need not be periodic, but we will mainly consider periodic examples

One-dimensional waves

- Waves depend on *two* variables: position ***and*** time
- For example, transverse displacement on a string, D, is a function of position and time



- Textbook refers to
  - *snapshot* graphs (D vs x for a given t as shown above)
  - *history* graphs (D vs t for a given x)



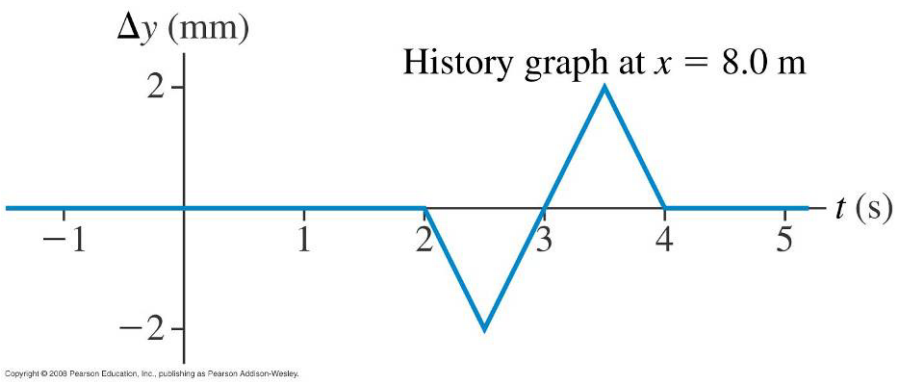


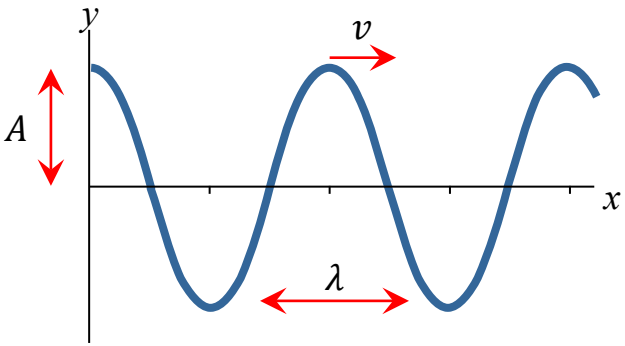
Figure from *Fundamentals of Physics* by Halliday Resnick and Walker (Wiley)

11.7 Some basic properties of a periodic travelling wave

- Amplitude A from zero to peak
- Wavelength λ from crest to crest
  … or trough to trough
  … or … (same phase in cycle)
- Period T from crest to crest etc.
- Velocity v speed at which crest (etc.) moves

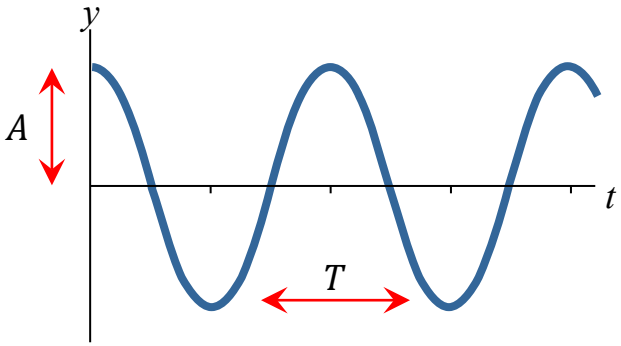
For *some instant in time* (snapshot):

$$y(x) = A\cos\left(\frac{2\pi}{\lambda}x\right)$$



For *some position in space* (history):

$$y(t) = A\cos\left(\frac{2\pi}{T}t\right)$$



D Wa-1 Wave Machines

Sinusoidal waves: $v = f\square$

- Velocity = distance per unit time
- Consider movement for exactly one period, T
- Must move one wavelength λ

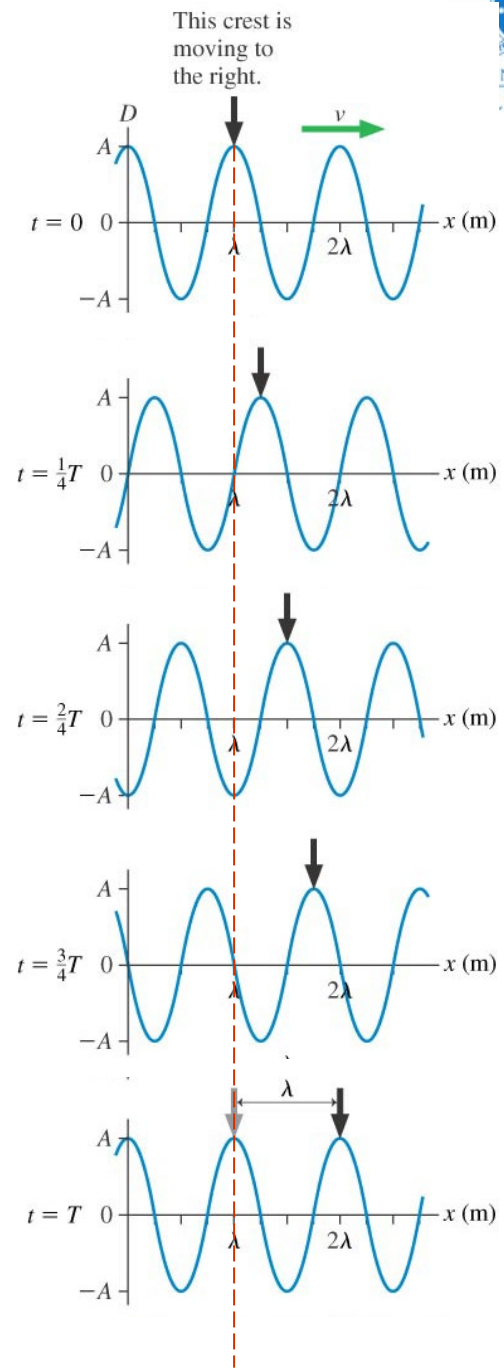
- Velocity is then $v = \frac{\lambda}{T} = f\lambda$

- That is: $\boxed{v = f\lambda}$

- "*THE fundamental relationship for sinusoidal waves*"

- Note: v is phase velocity: the speed of a point of constant phase. NOT the same as v_x for a SHM!

- Note:
  f is a property of the source,
  v is a property of the medium,
  λ therefore depends on both



(Extra background) Period and wavelength

E.g. sinusoidal wave travelling in the positive x direction:

$$D(x,t) = A\cos\left(2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right) = A\cos(kx - \omega t)$$

Period: time for one complete wave cycle at a given point

$$\cos(kx - \omega t) = \cos\big(kx - \omega(t + T)\big)$$

– True if $\omega T = 2\pi \Rightarrow$ $\boxed{T = \frac{2\pi}{\omega}}$

Wavelength: distance between repetitions at a given time

$$\cos(kx - \omega t) = \cos(k(x + \lambda) - \omega t)$$

– True if $k\lambda = 2\pi \Rightarrow$ $\boxed{\lambda = \frac{2\pi}{k}}$

k = "wavenumber" (**not** spring constant)

- Note $k \equiv \frac{2\pi}{\lambda}$ and $f \equiv \frac{\omega}{2\pi}$ and $v = \lambda f$ so $\boxed{\omega = vk}$

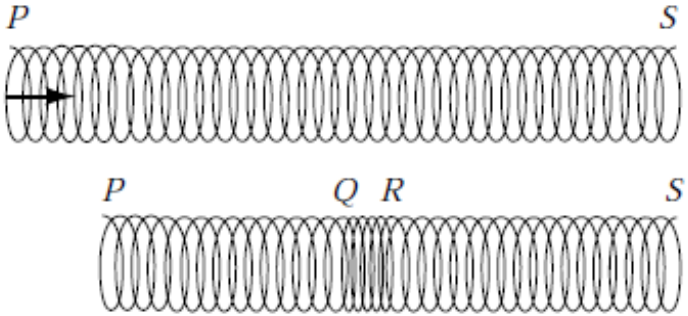
ConcepTest

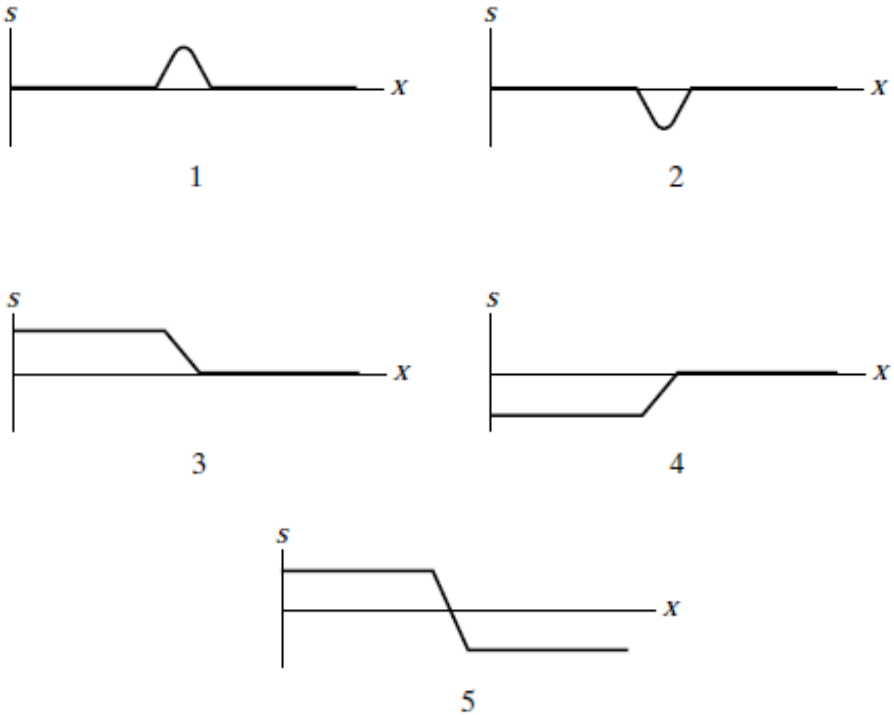
A wave is sent along a long spring by moving the left end rapidly to the right and keeping it there.

The figure shows the wave pulse at *QR*: part *RS* of the long spring is as yet undisturbed.

Which of the graphs 1–5 correctly shows the relation between displacement s and position x?

Displacements to the right are positive.





From Peer Instruction by E. Mazur

When poll is active, respond at **PollEv.com/martinsevior**

Longitudinal Wave

A wave is sent along a long spring by moving the left end rapidly to the right and keeping it there.

The figure shows the wave pulse at *QR*: part *RS* of the long spring is as yet undisturbed.

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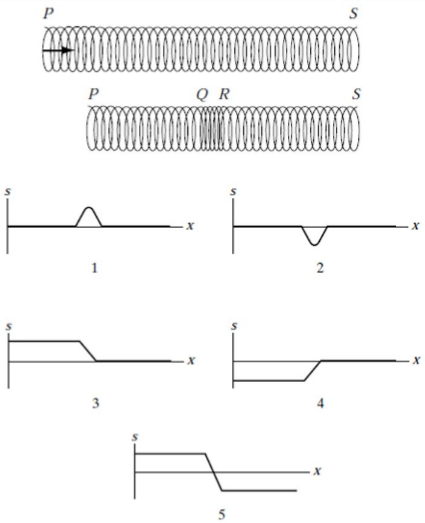

1.

2.

3.

4.

5.

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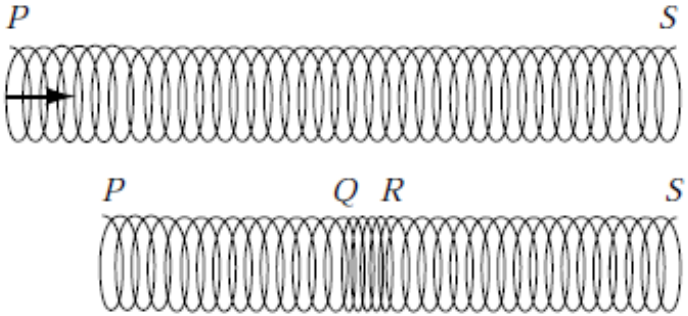
ConcepTest

A wave is sent along a long spring by moving the left end rapidly to the right and keeping it there.



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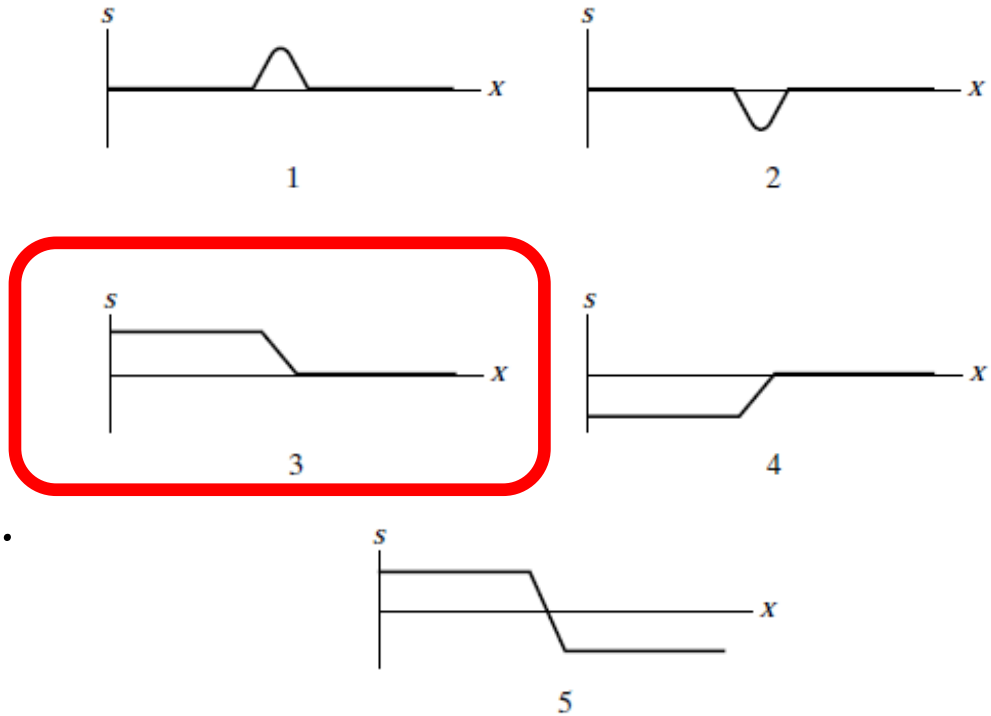
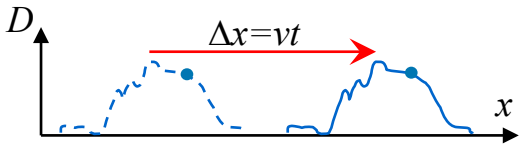


From Peer Instruction by E. Mazur

(Extra background) Phase velocity: $v=f\lambda$ again

- Consider a point of constant phase



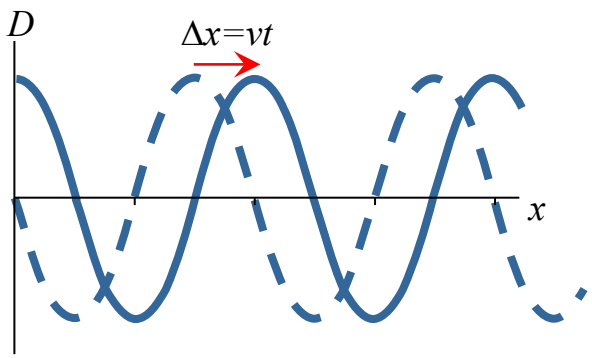
- *Phase velocity* is speed at which a point of constant phase travels

$$(kx - \omega t + \phi_0) = \text{constant}$$

$$\Rightarrow k\frac{dx}{dt} - \omega = 0$$

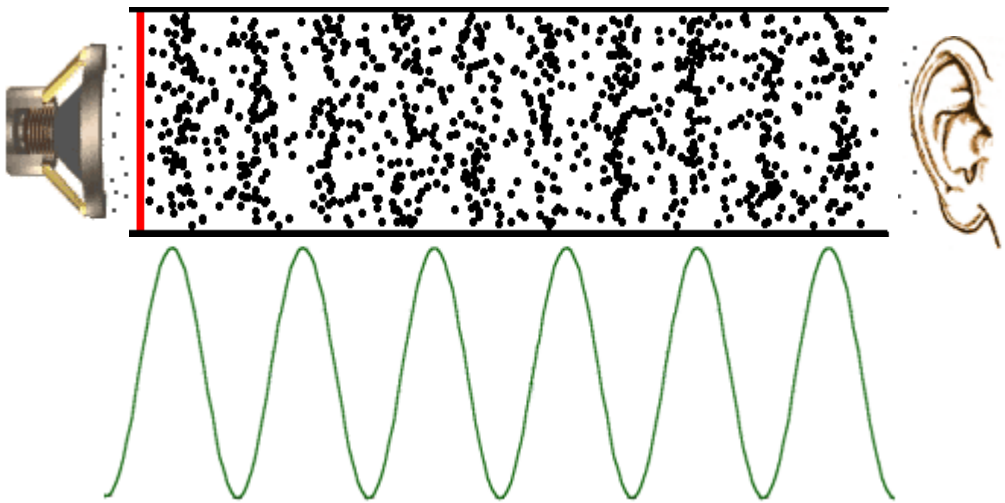
$$\Rightarrow \boxed{v = \frac{\omega}{k} \quad \text{or} \quad \omega = kv}$$

Phase velocity

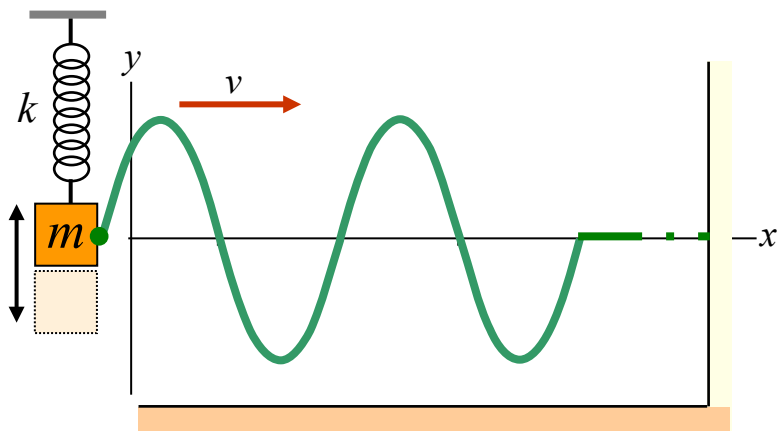


11.7 Sinusoidal waves

- A SHM oscillator can *generate* a travelling wave
- E.g. loudspeaker



- Spring and rope



12.1 SOUND WAVES: BASIC PROPERTIES

12.0 Sound waves

A pianist plays the note "middle C." The sound is made by the vibration of the piano string and is propagated outward as a vibration of the air (which can reach your ear). How does the vibration on the string compare to the vibration in the air?

(a) The vibration on the string and the vibration in the air have the same wavelength.

(b) They have the same frequency.

(c) They have the same speed.

(d) Neither wavelength, frequency, nor speed are the same in the air as on the string.

Giancoli p328

0:00

Respond at **PollEv.com/martinsevior**

ext **MARTINSEVIOR** to **+61 480 025 509** once to join, then **A, B, C, or D**

Soundwaves

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(a) The vibration on the string and the vibration in the air have the same wavelength. **A**

(b) They have the same frequency. **B**

(c) They have the same speed. **C**

(d) Neither wavelength, frequency, nor speed are the same in the air as on the string. **D**

Instructions Responses Clear responses

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12.0 Sound waves

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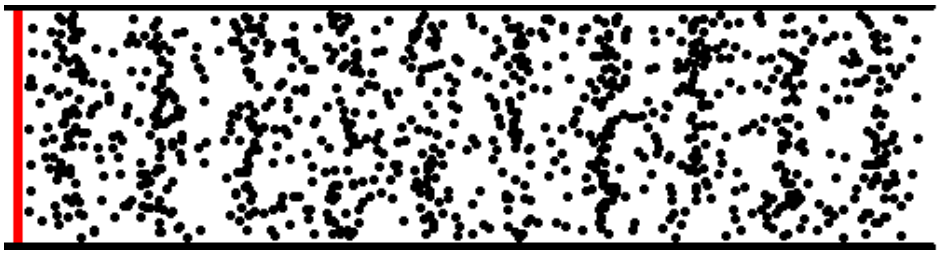
12.1 Sound waves

Sound is a longitudinal wave: regions of compression and expansion propagating in the direction of the pressure gradients



Speed of sound: depends on material. In air: $v \approx (331 + 0.60\,T\,)\ \mathrm{m/s}$ NB: T °C

Pitch: frequency; audible 20 Hz to 20 kHz

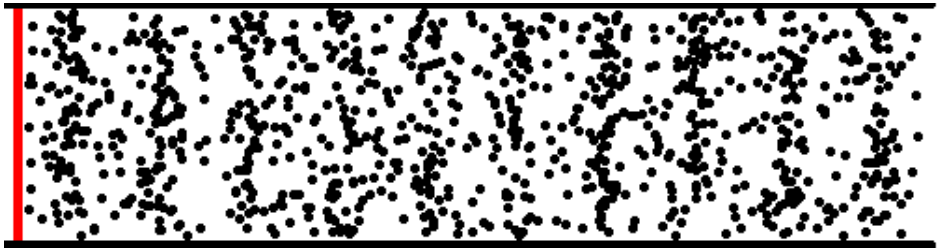
Wavelength: $\lambda = \frac{v}{f}$; audible ~ 20 m to 20 mm (1 kHz = 34 cm)

Loudness of sound: related to intensity (energy per unit time per unit area)

D Wa-9 Sound in a vacuum

12.1 Sound waves

- Sound is a longitudinal wave: regions of compression and expansion propagating in the direction of the pressure gradients



- Sound waves can be described in terms of displacement or pressure

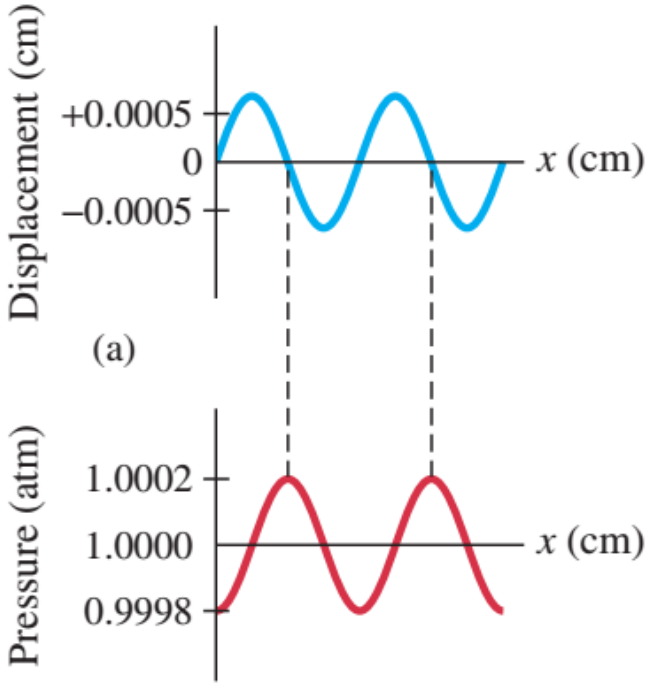
Displacement

- $s(x,t) = s_m \cos\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$

Pressure

- $p(x,t) = \Delta p_m \sin\left(\frac{2\pi}{\lambda}x - \frac{2\pi}{T}t\right)$
  - Note the phase shift!

- Typical values for $s_m, \Delta p_m$??



12.1 Sound waves: speed

- How fast?
- Expect speed dependent on:
  - Stiffness/elasticity
  - Density
- Can show: $v = \sqrt{\frac{B}{\rho}}$
  - B = bulk modulus (elasticity)
  - ρ = density

12.1 Speed of sound

- Speed of sound from material properties:

$$v = \sqrt{\frac{B}{\rho}}$$

B = "bulk modulus" (measure of how compressible)

ρ = mass density

Medium	Speed (m/s)
Air (0 ºC)	331
Air (20 ºC)	343
Helium	965
Water (20 ºC)	1482
Steel	5941
Granite	6000
Diamond	12000

Youtube demonstrations of change in sound velocity:

https://www.youtube.com/watch?v=d-XbjFn3aqE

https://youtu.be/FRKVQcbIByo

D **Wa-5 Sound in Helium**

Medium	Speed (m/s)
SF_6	133
Air (20 ºC)	343
Helium	965

https://www.youtube.com/watch?v=d-XbjFn3aqE https://youtu.be/FRKVQcbIByo

Example: Speed of Sound

Question: We see a lightning strike and later hear the thunder. Derive an expression for how far away the lightning is, in km, based on the time between seeing and hearing, in seconds. Use $v_{\text{air}} = 340$ m/s.

Solution:

We see the strike $t = \frac{d}{c} \approx 0$ seconds after it happens

We hear the strike $t = \frac{d}{v_{\text{air}}}$ seconds later

Hence $d = t\, v_{\text{air}}$

If we use v_{air} in km/s then $v_{\text{air}} \sim \frac{1}{3}$ km/s and the strike is $d = \frac{t}{3}$ km away

Sound Properties: 'Pitch' = frequency

Pitch

- Pitch = frequency
- Human sensation
- 'High pitch' (violin) or 'low pitch' (drum) sound

< 20 Hz	20 Hz to 20 kHz	> 20 kHz
Infrasonic	Audible	Ultrasonic
"feel" (sub-woofer)	Young human ear	Dog 50 kHz
		Bat 100 kHz
		Imaging 1 to 12 MHz

Wavelength

- Recall $v = f\lambda$. What is wavelength of sound?
  - $v_{\text{air}} \approx 340\ \text{m/s}$, $f = 20\ \text{Hz to } 20{,}000\ \text{Hz} \rightarrow \lambda \approx 17\ \text{m to } 17\ \text{cm}$
  - Vocal: $f \sim 500\ \text{Hz}$, $\lambda \sim 0.7\ \text{m}$

Example: Wavelength of Sound

Ultrasound medical imaging: Australian invention

Ultrasound imaging uses sound waves at high frequency, e.g. 7 MHz.

What is the wavelength of the sound wave in the human body? Use the velocity in water, $v \sim 1500$ m/s.

$v = f\lambda$ so $\lambda = v/f$ which gives $\lambda = 1500/(7 \times 10^6) = 0.2\ \mathrm{mm}$

Note that human hearing works best around $f = 500\ \mathrm{Hz}$ where the wavelength is $\lambda \sim 70\ \mathrm{cm}$ (similar to length of vocal tract)

1961: David Robinson and George Kossoff, Ultrasonic Research Group of the Commonwealth Acoustic Laboratories (CSIRO), built Australia's first ultrasound scanner, called the CAL Echoscope. While others had used the same ultrasound technology for similar purposes around the globe, it was discovered in 1962 that the results from the Australian version – called grey-scale ultrasound – were technically superior to other scanners, as well as being the first commercially practical option.

Summary

Waves

- Introduced "waves" as *propagating* disturbances $D\left(\frac{2\pi}{\lambda}x - \omega t\right)$
- Functions of <u>*two*</u> variables, e.g. x and t

$$D(x,t) = A\cos\left(2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right) = A\cos(kx - \omega t)$$

- Travelling waves; period, wavelength, *phase velocity* $\boxed{v = \lambda f}$